

# Class Imbalance Mitigation for TCGA-UT

## Abstract

Class imbalance in computational pathology depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. This report provides an overview and controlled comparison of representative imbalance-mitigation method classes on TCGA-UT, a typical long-tailed histopathology dataset. Two native-regime benchmarks are evaluated on shared slide-disjoint splits and three seeds. On the patch benchmark with frozen Virchow2 embeddings, combining cross-entropy with differentiable soft macro-F1 or soft Matthews-correlation objectives under balanced oversampling yields small but consistent gains over plain cross-entropy, whereas ProGAN augmentation and focal loss do not. On the WSI-bag benchmark, RankMix shows a clearer benefit, especially in balanced accuracy, with more modest macro-F1 gains and overlapping seed variability. The report therefore frames imbalance mitigation as a regime-dependent design choice rather than a universally transferable recipe.

## 1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive summaries. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware summaries as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

A common source of ambiguity in pathology imbalance studies is the input regime. Some methods are naturally image-level methods: they assume labeled images, image embeddings, or directly generated image samples. Other methods are naturally WSI-level methods: they assume weak labels over bags of instances and use attention, instance ranking, or bag-level representation learning. Comparing such methods in one pooled leaderboard can confound the imbalance mechanism with changes in supervision, input size, backbone, and aggregation.

TCGA-UT is useful for separating these questions because it supports two legitimate tasks with different native inputs. The dataset provides labeled histopathology patches cropped from pathologist-selected tumor regions and slide identities that can be grouped into WSI-level bags. Patch labels are class labels, not dense pixel masks or cell-level annotations. This report therefore uses the available supervision directly: patch-level methods are evaluated on controlled patch classification with frozen Virchow2 embeddings, while MIL methods are evaluated on slide-level feature bags built from the same foundation-model family.

The purpose is not to introduce a new mitigation method or to reproduce every cited method in all implementation detail. Instead, the main contribution is a benchmarking report that maps major mitigation classes onto a typical histopathology long-tail setting and compares their behavior under a shared protocol. The report makes three contributions: it separates patch-level and WSI-bag imbalance evaluation rather than mixing them in one leaderboard; it implements representative data-level, loss-level, hybrid, synthetic-generation, and MIL-specific interventions under fixed splits and frozen Virchow2 features; and it reports discrimination, classwise, calibration, and validation-tuning evidence so that weak fixed-protocol results can

be interpreted as either robust failures or possible under-tuning. Method fidelity is defined by implementing the defining imbalance mechanism in the regime for which that mechanism was designed, while holding the backbone family, split, input budget, and evaluation code fixed within each benchmark.

## 2 Dataset

TCGA-UT is derived from diagnostic *whole-slide images* (WSIs) in The Cancer Genome Atlas (TCGA), a pan-cancer resource that links molecular and clinical cancer data with large image collections (Tomczak et al., 2015). The TCGA-UT release turns this slide archive into a computational-pathology image dataset by extracting hematoxylin and eosin (H&E) tumor *patches* from pathologist-selected *uniform tumor regions*. It contains 1,608,060 RGB patches of size  $256 \times 256$  pixels from 8,736 diagnostic slides and 32 *cancer-type labels*. The patch labels are slide- or cancer-type labels, not dense tissue masks or cell-level annotations (Komura et al., 2022).

This structure makes TCGA-UT well suited for an imbalance study because the imbalance is native to the dataset rather than imposed only after sampling. The largest class is glioblastoma multiforme (GBM; 792 slides), while the smallest class is diffuse large B-cell lymphoma (DLBC; 28 slides). The resulting slide-level imbalance ratio is 28.3. As a consequence, the dataset exposes the practical tension that motivates this study: methods must learn from many visually related cancer categories while preserving performance on classes with far fewer slide-level examples.

| Property                    | Value                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------|
| Cancer-type labels          | 32                                                                                                                  |
| Diagnostic slides           | 8,736                                                                                                               |
| TCGA-UT patches             | 1,608,060 patches of $256 \times 256$ pixels                                                                        |
| Largest class               | Glioblastoma multiforme (GBM), 792 slides                                                                           |
| Smallest class              | Diffuse large B-cell lymphoma (DLBC), 28 slides                                                                     |
| Slide-level imbalance ratio | 28.3                                                                                                                |
| Shared split size per seed  | 6,116 train, 1,310 validation, 1,310 test slides                                                                    |
| Controlled patch input      | Resolution level 0; up to 30 deterministic patches per slide; one frozen Virchow2 embedding per patch (Section 2.2) |
| WSI-bag input               | Up to 30 deterministic frozen Virchow2 feature instances from the same slide identities                             |

Table 1: Dataset summary used by the shared patch and WSI-bag benchmarks.

### 2.1 Data Units and Terminology

Several data units are involved in the benchmark, and the distinction matters for method comparison. A *slide* or *WSI* is one diagnostic histopathology image associated with one cancer-type label. A *tumor region* is a pathologist-selected area of a slide from which TCGA-UT crops groups of patches. A *patch* is one  $256 \times 256$  RGB crop selected by the controlled manifest. A *patch embedding* is the frozen Virchow2 vector computed from that crop and consumed by the patch-level linear classifier. A *feature chunk* is a saved frozen-feature representation associated with a slide or slide subset. A *WSI bag* is the collection of frozen patch-feature instances used by the attention-MIL model for one slide-level prediction. *Head*, *body*, and *tail* classes refer to fixed support tiers derived from the full TCGA-UT slide distribution when reporting aggregate class-group metrics on the held-out test split. A *slide-disjoint split* means that all patches and features from one slide stay in exactly one of train, validation, or test. Table 10 in Appendix A lists the same terms in compact form for quick reference.

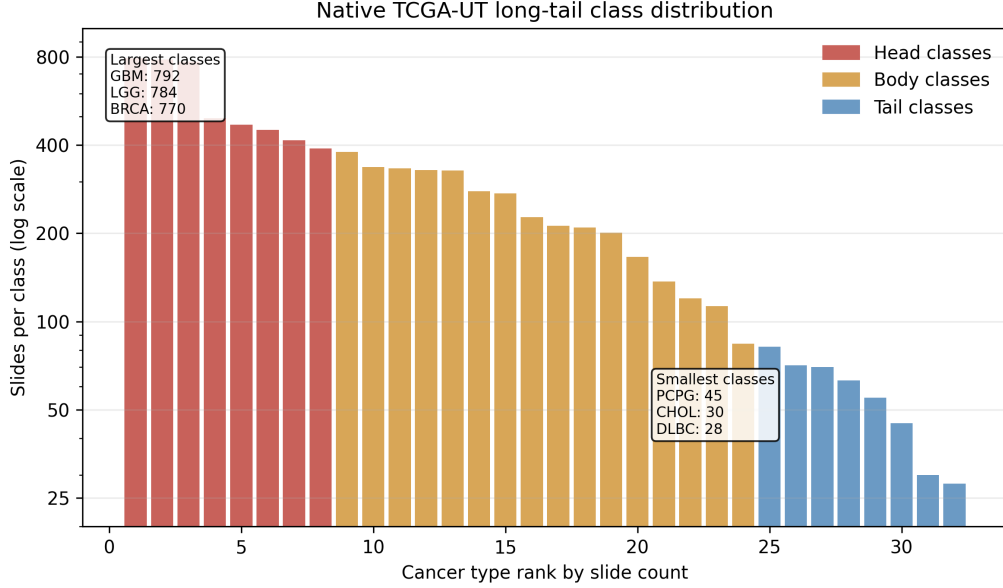
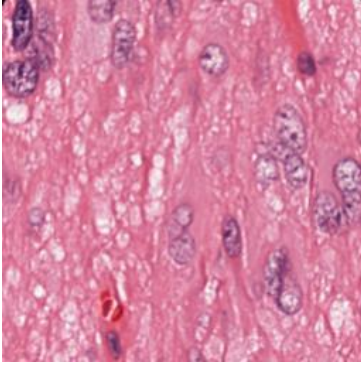


Figure 1: Native TCGA-UT slide support by cancer-type rank. The long-tailed distribution motivates metrics that weight classes more evenly than aggregate accuracy.



| Field            | Example value                                             |
|------------------|-----------------------------------------------------------|
| Cancer type      | Glioblastoma multiforme                                   |
| Resolution level | 0                                                         |
| Patch size       | 256 × 256 RGB pixels                                      |
| Source unit      | One crop from a selected tumor region                     |
| Patch role       | Image-level sample if selected by the controlled manifest |
| WSI-bag role     | Candidate instance in the slide’s feature bag             |

Table 2: Example TCGA-UT data point copied from the shared raw dataset. The image is a single patch, while the table shows how that patch relates to the manifest fields used by the benchmark.

## 2.2 Shared Splits and Benchmark Inputs

For each seed, slides are split into 6,116 training, 1,310 validation, and 1,310 test slides. The same slide-level assignments are reused by both benchmarks so that no patches from the same slide can appear in different splits. This is the main leakage-control mechanism: the patch benchmark and the WSI-bag benchmark see different input units, but they inherit the same train/validation/test slide partition.

The patch benchmark derives a deterministic controlled patch manifest from the shared slide split. It uses resolution level 0 and samples up to 30 patches per slide with a fixed rule before method-specific training begins. The cap is a compromise between data coverage and slide-level control: it uses the empirical median number of raw JPG patches per slide at resolution level 0 (median 30, mean 31.1, range 10–80) while preventing slides with many extracted tumor-region patches from dominating the patch-level loss. At this resolution, 8,686 of 8,736 slides have at least 30 raw patches, so the cap retains the typical per-slide patch evidence while limiting only the larger slide folders. Each selected patch is encoded once with a frozen Virchow2 model, and all patch methods train the same lightweight classifier head on these embeddings

(Zimmermann et al., 2024; Vorontsov et al., 2024). The encoder output is concatenated as  $\text{concat}(\text{CLS}, \text{mean}(\text{patch tokens}))$ , yielding 2,560-dimensional float16 vectors that are cast to float32 before classifier training. This design isolates imbalance mechanisms from both patch-selection policy and end-to-end image-backbone learning.

The WSI-bag benchmark applies the same per-slide evidence budget: when a frozen feature bag contains more than 30 instances, it is deterministically subsampled to 30 evenly spaced instances before attention-MIL training and evaluation. This keeps compute bounded and avoids a systematic advantage for slides with larger extracted feature bags. The two benchmarks still differ in supervision unit and aggregation: patch models classify each selected crop independently, whereas WSI-bag models predict one slide label from an attention-weighted set of up to 30 instances.

### 3 Methods

The methods are organized by the level at which they intervene in the imbalanced learning problem. Cross-entropy (CE) is treated as the no-mitigation reference condition: it uses the observed training distribution as given and does not rebalance samples, labels, costs, representations, or generated data. The remaining methods are tested imbalance interventions grouped according to the taxonomy used in the class-imbalance literature: data-level sampling and augmentation, algorithm-level losses, hybrid sampling-loss methods, synthetic generation, and representation or architecture-level methods (Saini and Susan, 2023; Zhang et al., 2023).

The comparison is regime-specific. Patch methods consume frozen Virchow2 embeddings of controlled patches and optimize one shared linear classifier head. WSI-bag methods consume capped frozen Virchow2 feature bags and optimize one shared attention-MIL head. Both benchmarks therefore rely on the same pathology foundation-model family, but they differ in supervision unit: one embedding per selected patch versus aggregation over a bounded set of instances on a slide. Method differences within a benchmark therefore correspond to imbalance mechanisms rather than changes in split, evaluation code, or encoder choice.

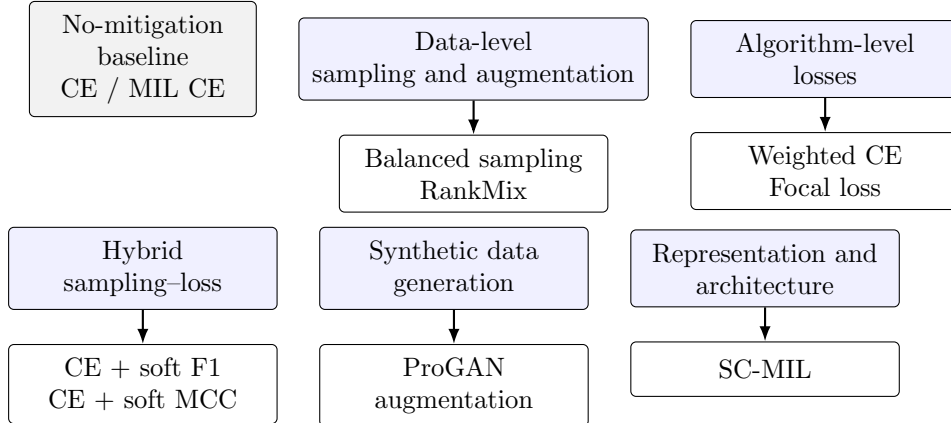


Figure 2: Method organization used in the benchmark. Cross-entropy is the no-mitigation reference, while the tested alternatives intervene through sampling, loss design, synthetic data, WSI feature augmentation, or representation learning.

#### 3.1 Notation

Let  $C$  be the number of cancer-type classes and let  $n_c$  denote the number of training examples from class  $c$ . For a labeled patch or slide-level bag with label  $y \in \{1, \dots, C\}$ , the model outputs

logits  $\mathbf{z} \in \mathbb{R}^C$  and class probabilities

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^C \exp(z_j)}. \quad (1)$$

For WSI-bag methods, a slide is represented as a bag  $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$  of frozen patch-feature instances. The attention-MIL backbone maps each bag to a bag embedding  $\mathbf{b}(B)$  and class logits. Unless a method explicitly changes the sampler, objective, or representation loss, the empirical class frequencies in the training set define the optimization signal.

### 3.2 Cross-Entropy Baseline

Unweighted cross-entropy is the baseline because it is the standard supervised objective without an explicit class-imbalance correction. For an example with label  $y$ , the loss is

$$\mathcal{L}_{\text{CE}}(\mathbf{z}, y) = -\log p_y. \quad (2)$$

In the patch benchmark, this objective trains the shared linear head on frozen Virchow2 patch embeddings from the controlled manifest. In the WSI-bag benchmark, the same objective trains the shared attention-MIL model on slide-level feature bags. In both cases, CE uses the observed class distribution directly: majority classes appear more often in minibatches and therefore contribute more loss terms over an epoch. This makes CE an appropriate reference for asking whether an imbalance-specific intervention actually improves the rare-class tradeoff.

### 3.3 Data-Level Sampling and Augmentation

Data-level methods change what the optimizer sees, rather than changing the mathematical penalty assigned to one observed example. In the taxonomy, this family includes resampling, augmentation, and WSI-specific feature mixing strategies that make minority or informative examples more visible during training (Saini and Susan, 2023).

#### 3.3.1 Balanced Sampling

Balanced sampling changes minibatch composition by sampling training examples with probabilities inversely related to their class support. For an example  $i$  with class  $y_i$ , the sampler weight is proportional to

$$q_i \propto \frac{1}{n_{y_i}}. \quad (3)$$

After normalization across the training set, this makes the expected class exposure closer to uniform even though the underlying dataset remains long-tailed. The patch variant applies this sampler to patch examples and still optimizes CE. The MIL variant applies the same idea at slide-bag level and also optimizes CE. Balanced sampling therefore tests a pure data-level intervention: it asks whether equalizing training exposure is enough without changing the loss formula.

#### 3.3.2 RankMix

RankMix is a WSI-native data-augmentation method for weakly supervised slide classification. Ordinary image mixup assumes two inputs with compatible spatial shapes, but WSI bags can contain different numbers of feature instances and lack instance-level labels. RankMix addresses this by mixing representative feature subsequences rather than raw slides (Chen and Lu, 2023).

The tested implementation follows the defining two-stage teacher–student procedure. First, a teacher attention-MIL model is trained with slide labels for 30 epochs using the same MIL

architecture and optimization settings as plain MIL CE; when available, the matched plain MIL CE checkpoint for the same seed is reused as the teacher. For each bag  $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$  and target class, the teacher assigns ranking scores to instances using its softmax probability for that class; the top- $k$  instances are kept as a representative subsequence, where  $k = \min(|B_a|, |B_b|)$  for the two bags being mixed, and their original within-bag order is preserved after selection. The student MIL model is then reinitialized and trained on bags built from these ranked subsequences.

Mixing itself follows the mixup idea in feature space. For two bags  $a$  and  $b$ , let  $H_a$  and  $H_b$  denote the selected instance-feature matrices and  $\mathbf{y}_a, \mathbf{y}_b$  the corresponding one-hot slide labels. A scalar mixing coefficient  $\lambda \in (0, 1)$  controls how much of each bag is retained:

$$\tilde{H} = \lambda H_a + (1 - \lambda) H_b, \quad \tilde{\mathbf{y}} = \lambda \mathbf{y}_a + (1 - \lambda) \mathbf{y}_b. \quad (4)$$

When  $\lambda$  is close to one, the mixed bag resembles bag  $a$ ; when it is close to zero, it resembles bag  $b$ . Values near  $\lambda = \frac{1}{2}$  interpolate more evenly between the two sources. RankMix does not fix  $\lambda$  to a single value. Instead, each mixed example draws

$$\lambda \sim \text{Beta}(\alpha, \alpha), \quad (5)$$

independently. The Beta distribution is a standard choice on the open interval  $(0, 1)$  because it is flexible and symmetric when both shape parameters are equal. With density

$$f(\lambda; \alpha, \alpha) \propto \lambda^{\alpha-1} (1 - \lambda)^{\alpha-1}, \quad \lambda \in (0, 1), \quad (6)$$

the parameter  $\alpha > 0$  controls how strongly mixing concentrates around the endpoints versus the center. For  $\alpha = 1$ ,  $\text{Beta}(1, 1)$  is the uniform distribution on  $(0, 1)$ , so every mixing weight is equally likely and the procedure reduces to ordinary symmetric mixup. Larger  $\alpha$  pushes  $\lambda$  toward  $\frac{1}{2}$ , producing more balanced interpolations; smaller  $\alpha$  pushes mass toward 0 and 1, producing mixtures that stay closer to one source bag. The benchmark fixes  $\alpha = 1.0$ , matching the default used in the RankMix reference implementation. The student is trained with soft-label cross-entropy on the mixed bags and labels  $\tilde{\mathbf{y}}$ . This mechanism is categorized as data-level augmentation because it increases the diversity of the training bags in feature space while retaining the same attention-MIL prediction architecture.

### 3.4 Algorithm-Level Losses

Algorithm-level methods keep the training examples fixed but alter how errors are penalized. They are cost-sensitive or difficulty-sensitive objectives: minority examples may receive larger explicit weights, or easy examples may contribute less to the gradient (Saini and Susan, 2023; Scholz et al., 2024).

#### 3.4.1 Class-Weighted Cross-Entropy

Class-weighted CE multiplies each example’s CE term by a class-dependent cost:

$$\mathcal{L}_{\text{WCE}}(\mathbf{z}, y) = -w_y \log p_y, \quad w_c \propto \frac{1}{n_c}. \quad (7)$$

The implementation normalizes the inverse-frequency weights so that the average scale of the loss remains comparable across methods. This method leaves the sample order and model architecture unchanged, but increases the loss contribution of classes with fewer training examples. It is evaluated for both patch classification and attention-MIL.

### 3.4.2 Focal Loss

Focal loss reduces the contribution of confidently classified examples and concentrates optimization on examples that remain difficult. With focusing parameter  $\gamma$ , the multiclass focal objective is

$$\mathcal{L}_{\text{focal}}(\mathbf{z}, y) = -(1 - p_y)^\gamma \log p_y. \quad (8)$$

This study uses  $\gamma = 2.0$  for the patch and WSI-bag variants. The method is relevant to imbalance because many minority-class examples can remain hard under a classifier dominated by head-class gradients, but the focal term does not directly encode the class counts. It is therefore a difficulty-sensitive loss rather than a sampler or explicit prior correction.

## 3.5 Hybrid Sampling–Loss Methods

Hybrid methods combine data-level and algorithm-level changes. The Scholz-style representatives are placed here because the tested variants use class-balanced oversampling together with metric-derived losses. This coupling matters: their effect cannot be attributed only to a new loss or only to a new sampler.

### 3.5.1 Scholz-Style CE + Soft F1

Scholz et al. motivate losses derived from metrics that are already used to evaluate imbalanced medical classification, including F1 and Matthews correlation coefficient (MCC) (Scholz et al., 2024). The intuition is that CE optimizes the log-probability of the correct class one example at a time, whereas macro F1 evaluates a different object: the balance between precision and recall at class level. In a long-tailed dataset, this distinction matters because a model can reduce CE substantially by fitting frequent classes while still leaving tail-class recall or precision weak.

The obstacle is that ordinary F1 is computed from hard counts after prediction, so it is not directly usable as a gradient-based training loss. The Scholz-style soft-F1 objective replaces those hard counts with probability-weighted counts inside each minibatch. For one class, a confident correct prediction contributes almost one soft true positive, a confident prediction for the wrong class contributes to soft false positives, and low probability on a true class contributes to soft false negatives. Formally, for one-hot labels  $y_i$  and predicted probabilities  $\hat{y}_i$ :

$$\text{TP} = \sum_i y_i \hat{y}_i, \quad \text{FP} = \sum_i (1 - y_i) \hat{y}_i, \quad \text{FN} = \sum_i y_i (1 - \hat{y}_i). \quad (9)$$

These terms keep the familiar meaning of a confusion matrix but make every component differentiable with respect to the model probabilities. They then define soft precision, soft recall, and soft F1:

$$\text{F1}_{\text{soft}} = \frac{2 \text{precision}_{\text{soft}} \text{recall}_{\text{soft}}}{\text{precision}_{\text{soft}} + \text{recall}_{\text{soft}}}. \quad (10)$$

The multiclass implementation applies this construction separately to each class and averages the resulting classwise terms. This macro-style averaging is the imbalance-aware part: each class contributes one term regardless of how many samples it has in the full dataset. The tested objective does not replace CE completely; it adds the metric-derived term to CE so that optimization still rewards calibrated class probabilities while also pushing the minibatch toward higher macro soft F1:

$$\mathcal{L}_{\text{CE+F1}} = \mathcal{L}_{\text{CE}} + \left( 1 - \frac{1}{C} \sum_{c=1}^C \text{F1}_{\text{soft},c} \right). \quad (11)$$

The benchmark trains this loss with balanced sampling on the patch manifest. That design is important for interpretation: the loss tries to optimize a macro-style class metric, while the sampler ensures that tail classes occur often enough in minibatches for their soft-F1 terms to provide a regular training signal.

### 3.5.2 Scholz-Style CE + Soft MCC

The soft-MCC variant uses the same principle as soft F1—replace hard evaluation counts with differentiable probability masses—but starts from the Matthews correlation coefficient (MCC). Whereas F1 for one class depends mainly on that class’s positive predictions, MCC is built from the full confusion matrix and therefore reacts to false positives, false negatives, and true negatives at the same time. In binary problems it is often preferred in imbalanced settings because it can remain informative when one class dominates the dataset: accuracy can look high while predictions are still systematically biased, but MCC penalizes both kinds of mistakes jointly (Scholz et al., 2024).

**Binary MCC as a correlation coefficient.** For one class treated as positive, write TP, TN, FP, and FN for the four confusion-matrix cells. The standard MCC is

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}. \quad (12)$$

The numerator compares co-occurrence of correct positive and correct negative decisions with co-occurrence of the two error types; the denominator scales by how many examples fall into each margin of the confusion matrix. MCC equals +1 for perfect predictions, 0 for chance-level association, and can become negative when predictions are worse than random. This makes it a correlation-style score rather than a prevalence-weighted rate.

**Multiclass MCC in a minibatch.** For  $C$  cancer types, Scholz et al. extend the idea to a minibatch of size  $N$  by treating labels and predictions as  $C$ -dimensional vectors per example (Scholz et al., 2024). Let  $Y \in \{0, 1\}^{N \times C}$  be the one-hot label matrix and  $\hat{Y} \in [0, 1]^{N \times C}$  the corresponding softmax probabilities, with rows  $\mathbf{y}_i$  and  $\hat{\mathbf{y}}_i$ . Three aggregate quantities describe how mass is distributed inside the batch:

$$s = \sum_{i=1}^N \sum_{c=1}^C Y_{ic} \hat{Y}_{ic}, \quad t_c = \sum_{i=1}^N Y_{ic}, \quad p_c = \sum_{i=1}^N \hat{Y}_{ic}. \quad (13)$$

Here  $s$  is the total probability-weighted agreement between labels and predictions: each example contributes  $\hat{y}_{i,y_i}$ , the predicted mass on its true class. The terms  $t_c$  and  $p_c$  are the batch totals of true and predicted mass for class  $c$ . They play the role of marginals in a confusion summary. If predictions collapse onto a few head classes, the vector  $(p_1, \dots, p_C)$  becomes skewed even when argmax accuracy looks acceptable; MCC is sensitive to that mismatch because it compares the joint alignment  $s$  with what would be expected from the marginals alone.

The multiclass MCC used in training can be written as a normalized covariance between the label matrix and the prediction matrix:

$$\text{MCC}_{\text{mc}} = \frac{Ns - \sum_{c=1}^C t_c p_c}{\sqrt{N^2 - \sum_{c=1}^C p_c^2} \sqrt{N^2 - \sum_{c=1}^C t_c^2}}. \quad (14)$$

The numerator  $Ns - \sum_c t_c p_c$  increases when true and predicted mass line up example by example and decreases when correct predictions are no better than what the marginal totals would suggest by chance. The two square-root terms measure how spread out the batch-level label totals and prediction totals are across classes; they prevent the score from exploding when one minibatch happens to contain only a few classes. Equation (14) reduces to the familiar binary form in (12) when  $C = 2$ .



**Soft MCC loss.** Hard MCC replaces  $\hat{Y}$  by one-hot argmax predictions, which again blocks gradient-based training. The soft-MCC objective keeps the structure of (14) but plugs in softmax probabilities directly:

$$\text{MCC}_{\text{soft}} = \frac{Ns - \sum_{c=1}^C t_c p_c}{\sqrt{N^2 - \sum_{c=1}^C p_c^2} \sqrt{N^2 - \sum_{c=1}^C t_c^2}}. \quad (15)$$

Every term in (13) and (15) is differentiable with respect to the logits that produce  $\hat{Y}$ . A confident correct prediction increases  $s$ ; probability placed on the wrong classes increases some  $p_c$  while leaving the matching  $t_c$  unchanged, which lowers the numerator; and overly peaked prediction vectors reduce the prediction-side denominator term, limiting the score even if a few examples are classified correctly. As with soft F1, the training objective turns a score to be maximized into a loss to be minimized and adds it to CE:

$$\mathcal{L}_{\text{CE+MCC}} = \mathcal{L}_{\text{CE}} + (1 - \text{MCC}_{\text{soft}}). \quad (16)$$

This variant is also evaluated only in the patch benchmark and is trained with class-balanced oversampling. The soft-F1 and soft-MCC variants therefore ask a focused question: once tail classes are regularly presented to the optimizer, does adding a differentiable version of a class-balanced evaluation metric improve the controlled patch classifier beyond CE and simple balanced sampling?

### 3.6 Synthetic Data Generation

Synthetic generation is data-level in spirit but is separated because it introduces new generated samples rather than resampling observed ones. In histopathology, this family is attractive because deleting majority-class tissue wastes data, while naive image-space interpolation often produces unrealistic pathology images (Saini and Susan, 2023; Ruiz-Casado et al., 2024). The practical hope is simple: if real minority-class patches are scarce, a generator may learn enough of their stain, texture, and tissue morphology distribution to provide additional plausible training examples.

#### 3.6.1 ProGAN Augmentation

The ProGAN experiment adapts the GAN-based histopathology augmentation idea of Ruiz-Casado et al. to multiclass TCGA-UT patches (Ruiz-Casado et al., 2024). A generative adversarial network (GAN) has two coupled models. The generator  $G$  maps random latent vectors to synthetic images, while the discriminator  $D$  tries to distinguish real training patches from generated patches. Training alternates between making  $D$  better at this distinction and making  $G$  produce images that  $D$  cannot reliably reject. The standard objective expresses this competition as

$$\min_G \max_D \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{u} \sim p(\mathbf{u})} [\log(1 - D(G(\mathbf{u})))] \quad (17)$$

This equation is not a classification loss for the final TCGA-UT classifier. It is the pretraining objective for producing additional patch images. After generation, the classifier simply sees a larger patch training set.

ProGAN modifies ordinary GAN training by growing the image resolution gradually. Instead of asking the generator to synthesize  $256 \times 256$  histology patches from the start, training begins at a coarse resolution where global color and texture structure is easier to model. Higher-resolution layers are then introduced step by step until the target patch size is reached. This staged procedure is intended to stabilize synthesis and reduce the risk that the generator learns only a narrow set of repeated images. In this benchmark, one class-specific generator is trained

for every training class whose patch count is below the head-class training count. Progressive training uses seven resolution stages from  $4 \times 4$  to  $256 \times 256$  pixels, with 25 epochs per stage, a fade-in fraction of 0.5 for newly added layers, latent dimension 128, base channel width 512, Adam optimization with learning rate  $2 \times 10^{-4}$  and  $\beta_1 = 0.5$ , and the depth-dependent batch schedule of Ruiz-Casado et al. When a class has more than 2,048 real training patches, at most 2,048 are drawn uniformly at random with a deterministic per-class seed for ProGAN training only; the downstream classifier still uses the full real training set together with the generated patches. Synthetic patch counts are set to the head-class training count minus the class training count, so each augmented tail class matches the largest real training class. Inception FID is computed for every augmented class between the generated images and the real training subset used for that class-specific generator; seed 0 has median FID 178.5, with values ranging from 103.0 to 315.9. No generated samples are filtered by FID or other quality thresholds. Real and synthetic RGB patches are embedded with the same frozen Virchow2 encoder before classifier training, so the augmentation is evaluated in the same embedding regime as the other patch methods. Synthetic images are used only as extra patch-classification training examples; they are not converted into synthetic WSI bags.

### 3.7 Representation and Architecture-Level Methods

Representation-level methods try to make rare classes more separable in feature space rather than only changing sampling or per-example loss weights. This family is especially relevant for computational pathology, where WSI labels supervise bags of many unlabeled instances and class imbalance can occur both across slides and within a positive slide (Juyal et al., 2024).

#### 3.7.1 SC-MIL

SC-MIL extends supervised contrastive learning to MIL by applying the contrastive objective to bag representations instead of assigning noisy slide labels to individual patches (Juyal et al., 2024). The motivation is that a slide label usually does not describe every patch in a bag. A cancer-type label can be correct for the slide while many instances contain background, normal tissue, artifacts, or weakly informative morphology. Applying contrastive learning directly to instances would therefore treat many patches as positives or negatives using labels they do not truly carry. SC-MIL avoids that mismatch by contrasting the aggregated slide-level representation.

The attention-MIL backbone first maps bag  $B_i$  to an embedding  $\mathbf{b}_i = \mathbf{b}(B_i)$ . A projection head  $g$  then produces normalized contrastive features  $\mathbf{r}_i = g(\mathbf{b}_i)$ . In the contrastive view, each bag in a minibatch is an anchor. Bags with the same cancer-type label are positives because their slide-level representations should become closer, while bags from other classes act as negatives because their representations should remain separated. For anchor  $i$ , bags of the same class form the positive set  $P(i)$ , and the supervised contrastive term is

$$\mathcal{L}_{\text{SCL}} = -\frac{1}{\sum_i |P(i)|} \sum_i \sum_{p \in P(i)} \log \frac{\exp(\mathbf{r}_i^\top \mathbf{r}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{r}_i^\top \mathbf{r}_a / \tau)}. \quad (18)$$

The numerator rewards similarity to same-class bags. The denominator compares that similarity against all other bags in the minibatch, so the loss favors representations where class clusters are compact and mutually separated. The temperature  $\tau$  controls how sharply these similarities are weighted; this benchmark uses  $\tau = 1.0$ .

SC-MIL does not rely on contrastive learning alone. The contrastive term shapes the representation space, while CE trains the final class predictor. The tested variant combines them through a linear curriculum:

$$\mathcal{L}_t = \beta_t \mathcal{L}_{\text{SCL}} + (1 - \beta_t) \mathcal{L}_{\text{CE}}, \quad \beta_t = 1 - \text{progress}_t. \quad (19)$$

Early training therefore emphasizes class-structured bag embeddings, when the classifier head is still unstable. As training progresses,  $\beta_t$  decreases and more weight moves to CE, so the learned representation is tied back to the actual slide-level classification objective. Activation diagnostics count positive pairs during training to confirm that the contrastive component is active.

### 3.8 Training Protocol and Reproducibility

All benchmark hyperparameters are fixed in the experiment configuration and are not tuned on the validation split. Validation metrics are written for inspection, but model selection, early stopping, and test reporting all use the weights after the final training epoch. Each benchmark is repeated over seeds 0, 1, and 2, which fix both the slide partition and the PyTorch random seeds used for initialization, shuffling, and sampling.

Table 3 summarizes the shared optimization settings. Patch and WSI-bag classifiers use AdamW with learning rate  $10^{-3}$ , weight decay  $10^{-4}$ , and 30 fixed epochs without early stopping. Patch training uses minibatches of 256 embeddings; WSI-bag training uses minibatches of 32 slide bags. Focal loss uses  $\gamma = 2.0$ , RankMix uses  $\alpha = 1.0$ , and SC-MIL uses contrastive temperature  $\tau = 1.0$ . These method-specific constants follow the cited references or common defaults rather than a validation sweep.

| Setting         | Patch benchmark | WSI-bag benchmark |
|-----------------|-----------------|-------------------|
| Optimizer       | AdamW           | AdamW             |
| Learning rate   | $10^{-3}$       | $10^{-3}$         |
| Weight decay    | $10^{-4}$       | $10^{-4}$         |
| Epochs          | 30              | 30                |
| Early stopping  | none            | none              |
| Minibatch size  | 256 patches     | 32 slide bags     |
| Model selection | final epoch     | final epoch       |
| Random seeds    | 0, 1, 2         | 0, 1, 2           |

Table 3: Shared training protocol for both benchmarks. All methods within a benchmark use the same optimizer, schedule, and epoch budget unless noted.

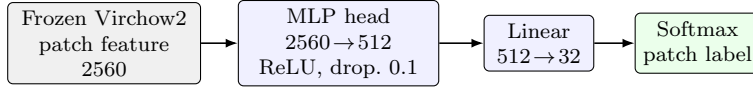
Figure 3 summarizes the two shared classifier heads used in the fixed protocol. Both benchmarks consume frozen 2,560-dimensional Virchow2 features and differ only in how those features are grouped before classification.

The patch head is a two-layer MLP: linear map to 512 units, ReLU, dropout rate 0.1, and a linear output to 32 classes. The WSI-bag head is an attention-based MIL network on the same feature dimension: each instance passes through the shared encoder, instance weights come from a linear scorer followed by a softmax over instances, the bag representation is their weighted sum, and slide logits are produced by a final linear map. SC-MIL additionally routes bag embeddings through a two-layer projection head of width 256 with ReLU. The WSI-bag input is capped at 30 instances per slide using deterministic evenly spaced subsampling.

RankMix uses balanced slide-bag sampling during student training. Mixed bags are formed within each minibatch by pairing bags with a random permutation and mixing their teacher-ranked representative subsequences with  $\lambda \sim \text{Beta}(\alpha, \alpha)$  and  $\alpha = 1.0$  as in (5). The student is reinitialized after teacher preparation and trained for the same 30-epoch budget as the other MIL methods.

Head, body, and tail summaries group classes by support in the full TCGA-UT slide distribution rather than by held-out test counts. Cancer types are sorted by dataset slide count; the eight lowest-support classes form the tail tier, the eight highest-support classes form the head tier, and the remaining sixteen classes form the body tier. Tier-wise precision, recall, and F1 on validation and test splits then aggregate only over the predefined classes in each tier.

(a) Patch benchmark



(b) WSI-bag benchmark

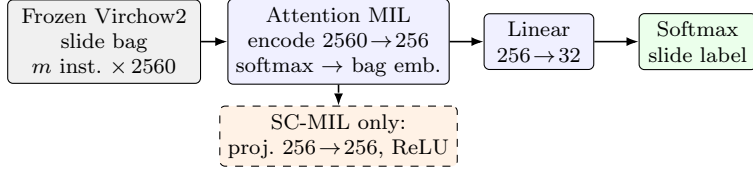


Figure 3: Shared classifier heads for the fixed training protocol. Panel (a) classifies one controlled patch embedding per forward pass. Panel (b) encodes all instances on a slide, pools them with attention, and predicts one slide-level label; the dashed branch is used only by SC-MIL. Virchow2 features are frozen in both benchmarks.

## 4 Evaluation

Both benchmarks use seeds 0, 1, and 2. Macro F1 is the primary ranking metric within each benchmark because it weights classes equally and penalizes failure on rare classes. Balanced accuracy, accuracy, classwise precision/recall/F1, and head/body/tail summaries are reported in the main result tables using the tier definitions in Section 3.8. These discrimination metrics answer whether the predicted class is usually correct, but they do not by themselves say whether the reported class probabilities are trustworthy. A model can rank well on macro F1 while assigning near-certainty to many wrong predictions, which is especially relevant in imbalanced medical classification where deployment and thresholding may depend on confidence rather than only on the argmax label (Mosquera et al., 2024). The study therefore reports three complementary probability-quality metrics on held-out softmax outputs; their definitions are given below and their values are summarized in Table 9. Patch and WSI-bag methods are never ranked against each other, because they answer related but different questions with different input units and supervision.

All reported tables use the held-out test split and show mean  $\pm$  standard deviation over seeds. Table 7 additionally reports paired seed-wise differences for the headline comparisons against each regime’s CE baseline; bracketed values list deltas in seed order 0–2 without formal significance testing. Separate validation summaries are retained as artifacts for model-selection checks. The completed sweep contains no missing planned results for either benchmark.

### 4.1 Probability Quality and Calibration

Let  $N$  denote the number of held-out examples in one benchmark split,  $C = 32$  the number of cancer types,  $y_i \in \{1, \dots, C\}$  the true label, and  $\hat{\mathbf{p}}_i \in [0, 1]^C$  the corresponding softmax probability vector with  $\sum_c \hat{p}_{ic} = 1$ . Write  $Y_i \in \{0, 1\}^C$  for the one-hot encoding of  $y_i$ . All three metrics below are computed from the raw model probabilities at the final training epoch, without temperature scaling or other post-hoc calibration.

**Negative log likelihood.** Cross-entropy training optimizes the log-probability of the correct class. A natural held-out analogue is the average negative log likelihood (NLL), also called

multiclass log loss:

$$\text{NLL} = -\frac{1}{N} \sum_{i=1}^N \log \hat{p}_{i,y_i}. \quad (20)$$

Lower NLL means the model assigns higher probability mass to the true class on average. NLL therefore measures both whether the top prediction is correct and how sharply probability is placed on that class: a correct but overconfident model and a correct but cautious model can have different NLL even when their hard-error rates match. In implementation,  $\hat{p}_{i,y_i}$  is clipped to  $[10^{-12}, 1]$  before taking the logarithm so that numerical underflow does not dominate the score when a class receives extremely small probability.

**Multiclass Brier score.** The Brier score was introduced for probabilistic forecasts and penalizes the entire predicted distribution rather than only the argmax label (Brier, 1950). For multiclass problems it is the mean squared error between  $\hat{\mathbf{p}}_i$  and  $Y_i$ :

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C (\hat{p}_{ic} - \mathbb{I}[y_i = c])^2 = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{p}}_i - Y_i\|_2^2. \quad (21)$$

Lower BS is better. The score is minimized when  $\hat{\mathbf{p}}_i$  equals the true class indicator and increases when probability leaks into incorrect classes. Unlike NLL, BS is bounded and symmetric in probability errors across classes, so it is less dominated by a single very small  $\hat{p}_{i,y_i}$ . Together, NLL and BS summarize overall distributional fit on the test split.

**Expected calibration error.** Discrimination and distributional fit can still leave confidence misaligned with empirical accuracy. Expected calibration error (ECE) measures that gap using confidence bins (Naeini et al., 2015; Guo et al., 2017). For each example define the predicted class  $\hat{y}_i = \arg \max_c \hat{p}_{ic}$  and the confidence  $c_i = \max_c \hat{p}_{ic}$ . Partition examples into  $B = 10$  equal-width bins on  $[0, 1]$  with edges  $\tau_0 = 0 < \tau_1 < \dots < \tau_{10} = 1$  and intervals  $(\tau_{b-1}, \tau_b]$ . Let  $B_b = \{i : c_i \in (\tau_{b-1}, \tau_b]\}$ . The bin accuracy and mean confidence are

$$\text{acc}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} \mathbb{I}[\hat{y}_i = y_i], \quad \bar{c}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} c_i. \quad (22)$$

ECE is the size-weighted average absolute gap between these quantities:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \bar{c}(B_b)|. \quad (23)$$

Lower ECE is better. Intuitively, if a model reports  $c_i \approx 0.9$  on many examples, then roughly 90% of those examples should be correct; large systematic deviations indicate over- or under-confidence. ECE is a coarse summary—it depends on binning and uses only the maximum predicted probability—but it is useful here because it separates confidence alignment from headline discrimination and highlights cases where a method improves macro F1 without improving the reliability of its reported probabilities.

## 5 Results

Both benchmarks share slide-disjoint splits, the Virchow2 feature family, and a maximum of 30 feature instances per slide, but they answer different prediction questions. The patch benchmark treats each selected crop as an independent labeled example; the WSI-bag benchmark predicts one slide label from an attention-weighted aggregation over the selected instances on that slide. Patch metrics therefore characterize patch-level classification only and must not be read as slide-level diagnostics; WSI-bag metrics characterize slide-level MIL only. Absolute scores and method orderings are not comparable across the two tables.

## 5.1 Patch-Level Results

| Method                   | Accuracy          | Balanced accuracy | Macro F1          |
|--------------------------|-------------------|-------------------|-------------------|
| CE + soft MCC (balanced) | $0.837 \pm 0.007$ | $0.782 \pm 0.008$ | $0.782 \pm 0.007$ |
| CE + soft F1 (balanced)  | $0.837 \pm 0.004$ | $0.782 \pm 0.004$ | $0.782 \pm 0.004$ |
| Weighted CE              | $0.830 \pm 0.004$ | $0.774 \pm 0.006$ | $0.776 \pm 0.012$ |
| CE                       | $0.834 \pm 0.002$ | $0.770 \pm 0.003$ | $0.776 \pm 0.002$ |
| Balanced sampler         | $0.832 \pm 0.005$ | $0.777 \pm 0.009$ | $0.775 \pm 0.008$ |
| ProGAN augmentation      | $0.830 \pm 0.003$ | $0.767 \pm 0.011$ | $0.774 \pm 0.008$ |
| Focal                    | $0.830 \pm 0.004$ | $0.760 \pm 0.002$ | $0.766 \pm 0.003$ |

Table 4: Patch-level test results over three seeds. Methods share the same controlled manifest, frozen Virchow2 encoder, linear classifier head, and evaluation code.

The patch benchmark compresses into a narrow performance band once patches are represented with frozen Virchow2 embeddings (Table 4). The Scholz-style hybrid objectives rank highest, but their advantage over plain CE is small: paired macro-F1 differences are positive in all three seeds, while the aggregate gaps remain within a few thousandths (Table 7). This pattern suggests that optimizing a class-balanced surrogate for macro-level behavior helps at the decision-boundary level, but does not create a qualitatively different patch classifier.

The simpler patch interventions separate mainly by what they fail to improve under this fixed protocol. Class-weighted CE is effectively tied with CE on mean macro F1, balanced sampling is slightly lower on macro F1 despite similar balanced accuracy, and ProGAN augmentation stays within the non-generative spread despite adding a large synthetic tail-class volume (Table 5). The accompanying FID and Virchow2 nearest-neighbor diagnostics indicate that this is not merely a quantity problem: many generated patches are visually degenerate or near-monochrome and are distant from the real feature manifold used by the classifier (Figure 4). Focal loss is weakest on macro F1, consistent with the interpretation that down-weighting easy examples is not helpful when most cancer types are already separable in the frozen embedding space.

| Class                                           | Real train | Generated | Inception FID | Virchow2 mean NN |
|-------------------------------------------------|------------|-----------|---------------|------------------|
| Lymphoid Neoplasm Diffuse Large B-cell Lymphoma | 600        | 16,000    | 315.9         | 52.16            |
| Cholangiocarcinoma                              | 660        | 15,940    | 232.2         | 49.09            |
| Pheochromocytoma and Paraganglioma              | 930        | 15,670    | 301.3         | 52.59            |
| Uveal Melanoma                                  | 1,160      | 15,440    | 255.9         | 52.46            |
| Rectum adenocarcinoma                           | 1,350      | 15,250    | 178.5         | 51.31            |
| Uterine Carcinosarcoma                          | 1,460      | 15,140    | 161.8         | 49.41            |
| Mesothelioma                                    | 1,500      | 15,100    | 249.1         | 46.97            |
| Kidney Chromophobe                              | 1,740      | 14,860    | 207.7         | 40.87            |
| All augmented classes (31 total)                | 166,480    | 348,120   | 178.5 median  | 47.89 median     |

Table 5: ProGAN augmentation diagnostics for seed 0. “Generated” counts are synthetic patches added per class to reach the head-class training patch count. Inception FID compares generated patches to the real training subset used for GAN training. Virchow2 mean NN is the mean, over a fixed synthetic subsample, of the minimum  $L_2$  distance to real same-class embeddings in the frozen feature cache.

The classwise-recall plot (Figure 6) shows that remaining errors are distributed across many cancer types rather than concentrated in a single tail class. In this setting, the main practical

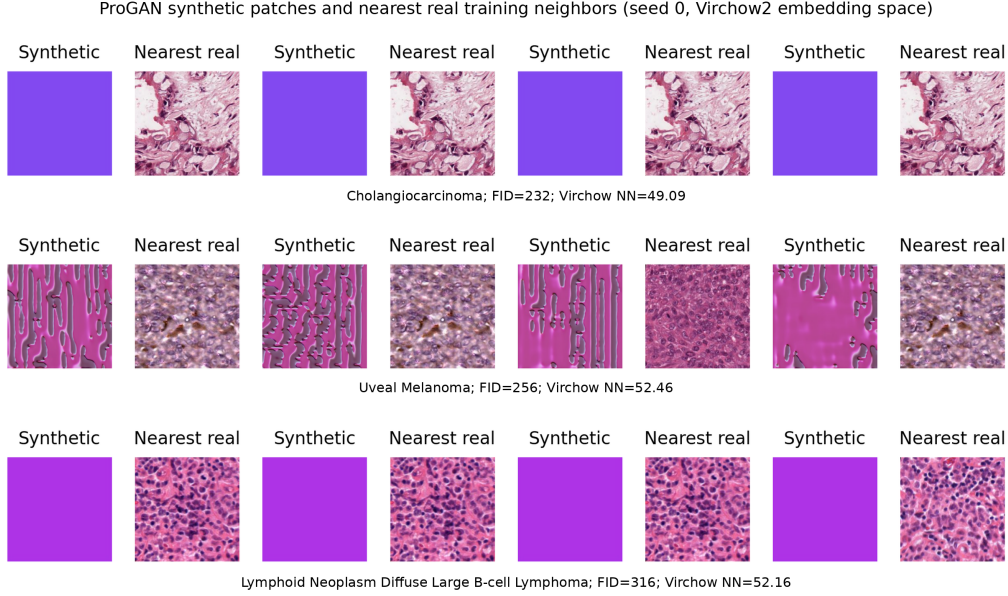


Figure 4: ProGAN synthetic patches (left column of each pair) and their nearest real training neighbors in Virchow2 embedding space (right column) for three tail classes in seed 0. The near-monochrome and visibly unrealistic synthetic patches are diagnostic failure cases rather than plausible histology examples; they explain why ProGAN augmentation is interpreted together with FID, nearest-neighbor distance, and classifier performance.

distinction is therefore not whether imbalance mitigation works at all, but which mechanism best preserves rare-class recall when the feature extractor is already strong. These conclusions remain confined to the patch benchmark: high patch accuracy or macro F1 does not directly establish slide-level utility, because each test prediction is made on a single crop rather than by aggregating evidence across a slide bag.

## 5.2 WSI-Bag Results

| Method       | Accuracy          | Balanced accuracy | Macro F1          |
|--------------|-------------------|-------------------|-------------------|
| RankMix      | $0.894 \pm 0.010$ | $0.863 \pm 0.008$ | $0.848 \pm 0.017$ |
| MIL CE       | $0.890 \pm 0.006$ | $0.836 \pm 0.020$ | $0.840 \pm 0.020$ |
| Balanced MIL | $0.885 \pm 0.007$ | $0.837 \pm 0.015$ | $0.836 \pm 0.013$ |
| SC-MIL       | $0.881 \pm 0.007$ | $0.837 \pm 0.017$ | $0.833 \pm 0.016$ |
| Weighted MIL | $0.880 \pm 0.005$ | $0.834 \pm 0.018$ | $0.829 \pm 0.018$ |
| Focal MIL    | $0.879 \pm 0.004$ | $0.824 \pm 0.011$ | $0.827 \pm 0.009$ |

Table 6: WSI-bag test results over three seeds. Methods share the same capped feature bags, attention-MIL backbone family, and evaluation code.

The WSI-bag benchmark evaluates one slide-level prediction per diagnostic slide by aggregating capped feature bags from the shared split. The 30-instance cap aligns the per-slide evidence budget with the controlled patch manifest and removes the earlier possibility that larger slides benefit simply from contributing more instances. The remaining difference is architectural and supervisory: MIL learns slide-level attention over a set of instances, whereas the patch classifier treats each crop as an independently labeled example. Table 6 should be read

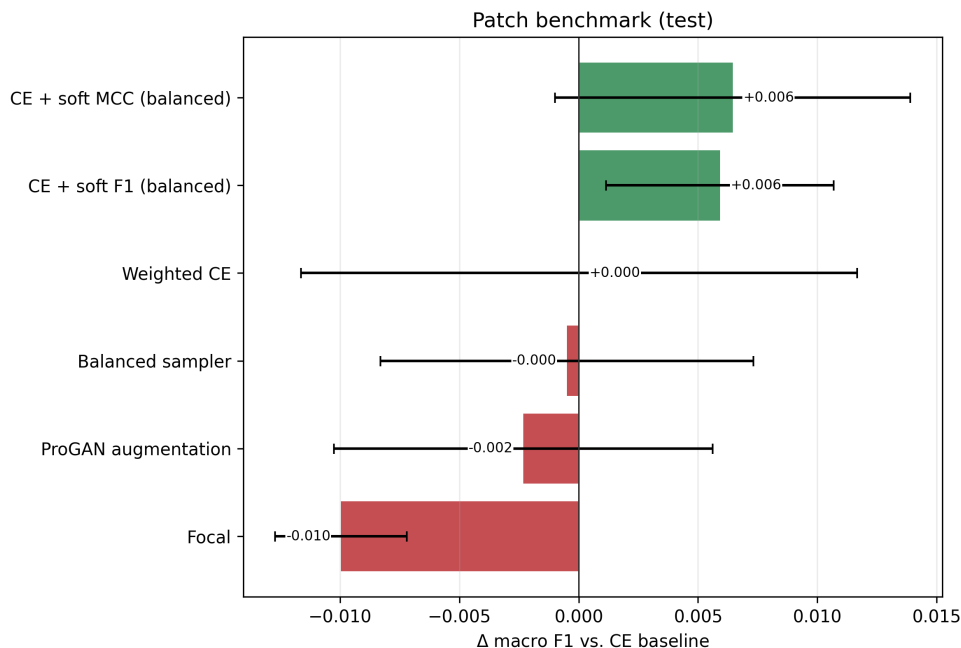


Figure 5: Patch-level change in macro F1 relative to the CE baseline on the held-out test split. Bars show the mean  $\Delta$  over seeds 0–2 with seed-wise standard deviation. Positive values indicate improvement over CE; near-zero values indicate no meaningful mean change. Values should only be interpreted within the patch benchmark (Table 4).

as a within-regime slide-level leaderboard, not as confirmation or refutation of the patch-level ranking.

RankMix shows the clearest WSI-bag benefit, especially in balanced accuracy, while macro-F1 gains remain modest (Table 6). Paired seed-wise comparisons show positive RankMix deltas in all three seeds for both headline metrics (Table 7), and calibration improves more clearly than discrimination (Table 9). This is consistent with RankMix acting on the MIL-specific bottleneck: it augments representative instance subsequences rather than only reweighting already observed bags.

The remaining WSI methods cluster closely. Balanced-sampling MIL and SC-MIL produce similar balanced accuracy (0.837), but both trail RankMix in macro F1. SC-MIL performs well on validation but does not exceed plain MIL CE on the test macro-F1 average. Weighted MIL CE and focal MIL do not improve over plain MIL CE in this configuration, with focal MIL giving the lowest WSI macro F1. Activation diagnostics confirm that the advanced WSI objectives were active: RankMix generated about 1,100,000 mixed examples per seed, and SC-MIL constructed about 1,060,000 positive pairs per seed.

### 5.3 Paired Seed Comparisons

Table 7 makes the headline gains explicit at seed resolution rather than only through aggregate means. Both patch hybrid objectives remain above plain CE on macro F1 in every seed, and RankMix remains above plain MIL CE on both metrics in every seed, even though the absolute shifts are modest and would require formal paired testing to treat as significant.

### 5.4 Validation-Tuned Robustness Check

The fixed-protocol results do not by themselves distinguish a genuinely ineffective mitigation mechanism from an under-tuned one. A targeted robustness check therefore tunes the



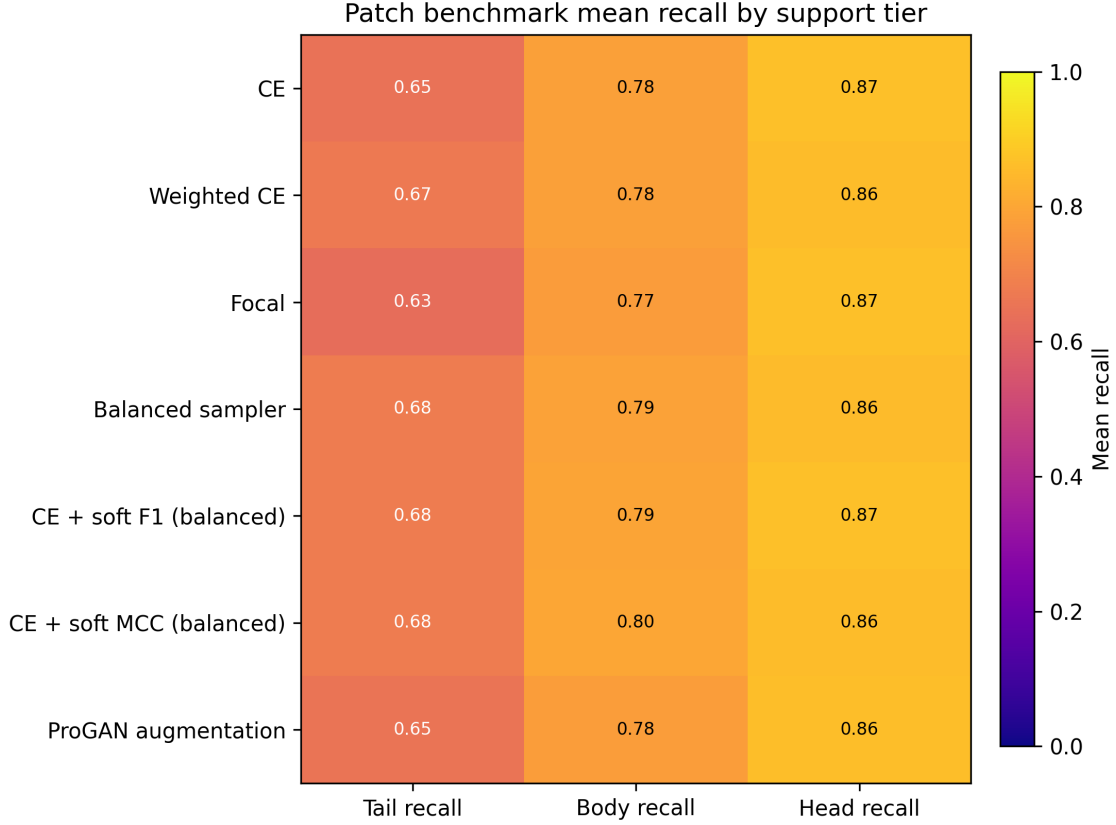


Figure 6: Patch-level classwise recall on the held-out test split. The plot exposes the class-level variation hidden by aggregate accuracy and motivates reporting balanced accuracy and macro F1.

imbalance-specific knobs of the non-ProGAN mitigation methods on validation macro F1, using balanced accuracy only as a tie-breaker, and then reports the selected configurations on the held-out test split. The tuning grid covers class-weight strength, sampler strength, focal-loss  $\gamma$ , Scholz metric-loss weight, RankMix mixing strength, and SC-MIL temperature; ProGAN is excluded because the generated images show clear quality failures rather than a classifier-only hyperparameter issue.

The tuned robustness check partly narrows the negative result but does not overturn it. On patches, validation tuning gives small test gains for weighted CE and balanced sampling, yet the two Scholz-style hybrid objectives remain the strongest patch methods; soft MCC selects its original metric-loss weight, and soft F1 improves only marginally with a larger weight. On WSI bags, RankMix is the only tuned method that clearly exceeds plain MIL CE on held-out macro F1, with  $\alpha = 2.0$  improving both macro F1 and balanced accuracy. Weighted CE,

| Comparison         | $\Delta$ Macro F1                           | $\Delta$ Balanced accuracy                  |
|--------------------|---------------------------------------------|---------------------------------------------|
| CE + soft MCC – CE | $+0.006 \pm 0.007$ [+0.001, +0.017, +0.001] | $+0.012 \pm 0.006$ [+0.005, +0.020, +0.011] |
| CE + soft F1 – CE  | $+0.006 \pm 0.005$ [+0.005, +0.012, +0.001] | $+0.012 \pm 0.001$ [+0.013, +0.013, +0.011] |
| RankMix – MIL CE   | $+0.008 \pm 0.006$ [+0.017, +0.005, +0.003] | $+0.027 \pm 0.013$ [+0.045, +0.015, +0.020] |

Table 7: Paired test-split differences against the within-regime CE baseline on seeds 0–2. Each cell reports mean  $\pm$  SD across seeds; bracketed values list seed-ordered paired deltas. No formal  $p$ -values are computed.

| Regime  | Method                   | Selected parameters    | Fixed test F1 | Val F1 | Tuned test F1 | Tuned bal. acc. |
|---------|--------------------------|------------------------|---------------|--------|---------------|-----------------|
| Patch   | CE                       | fixed baseline         | 0.776         | –      | 0.776         | 0.770           |
| Patch   | Weighted CE              | weight_power=0.75      | 0.776         | 0.793  | 0.779         | 0.778           |
| Patch   | Focal                    | focal_gamma=1          | 0.766         | 0.781  | 0.773         | 0.770           |
| Patch   | Balanced sampler         | sampler_power=0.75     | 0.775         | 0.792  | 0.780         | 0.774           |
| Patch   | CE + soft F1 (balanced)  | metric_loss_weight=20  | 0.782         | 0.797  | 0.782         | 0.780           |
| Patch   | CE + soft MCC (balanced) | metric_loss_weight=10  | 0.782         | 0.795  | 0.782         | 0.782           |
| WSI bag | MIL CE                   | fixed baseline         | 0.840         | –      | 0.840         | 0.836           |
| WSI bag | MIL weighted CE          | weight_power=0.25      | 0.849         | 0.848  | 0.835         | 0.836           |
| WSI bag | MIL focal                | focal_gamma=1          | 0.827         | 0.839  | 0.828         | 0.824           |
| WSI bag | MIL balanced sampler     | sampler_power=0.75     | 0.836         | 0.844  | 0.839         | 0.842           |
| WSI bag | RankMix                  | rankmix_alpha=2        | 0.848         | 0.854  | 0.856         | 0.866           |
| WSI bag | SC-MIL                   | sc_mil_temperature=0.8 | 0.838         | 0.848  | 0.838         | 0.838           |

Table 8: Validation-selected tuning robustness check. Configurations are selected only from validation macro F1 averaged over seeds 0–2; held-out test metrics are reported after selection and do not influence the choice.

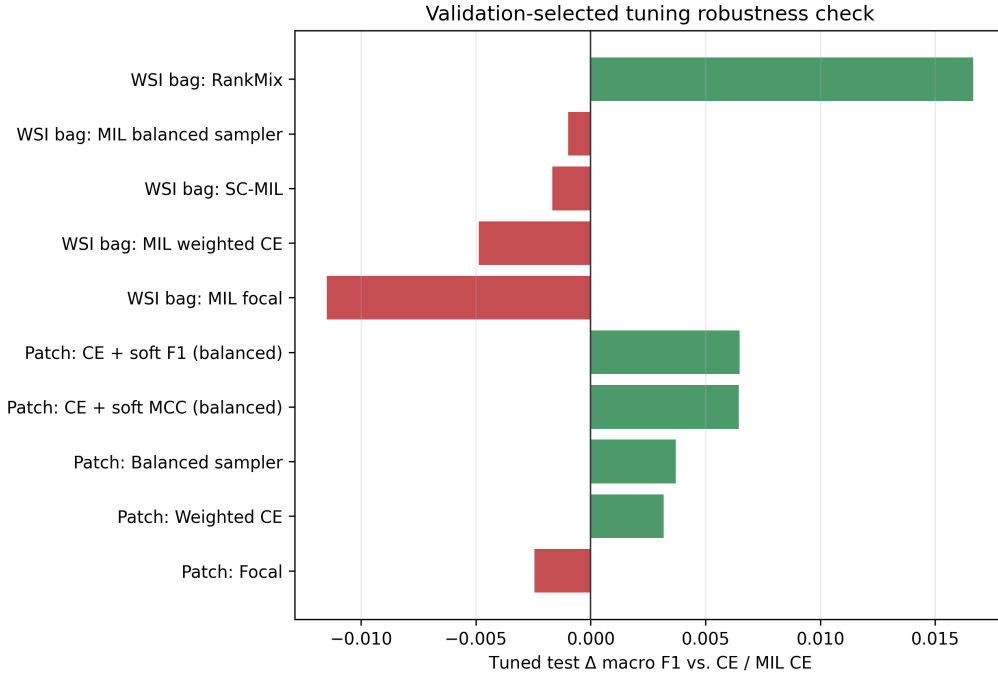


Figure 7: Held-out test macro-F1 change of validation-selected tuned configurations relative to the fixed CE or MIL CE baseline.

balanced sampling, and SC-MIL improve relative to their fixed settings but still do not surpass MIL CE on test macro F1, and focal loss remains below baseline. The fixed-feature, strong-baseline bottleneck is therefore still supported, although the tuned results show that some simple mitigation variants were mildly under-tuned.

## 5.5 Probability Calibration

| Regime  | Method                   | NLL               | Brier             | ECE               |
|---------|--------------------------|-------------------|-------------------|-------------------|
| Patch   | CE + soft MCC (balanced) | $1.952 \pm 0.092$ | $0.296 \pm 0.012$ | $0.140 \pm 0.006$ |
| Patch   | CE + soft F1 (balanced)  | $2.202 \pm 0.089$ | $0.301 \pm 0.008$ | $0.144 \pm 0.004$ |
| Patch   | Weighted CE              | $1.716 \pm 0.003$ | $0.302 \pm 0.007$ | $0.140 \pm 0.003$ |
| Patch   | CE                       | $1.808 \pm 0.087$ | $0.299 \pm 0.005$ | $0.141 \pm 0.003$ |
| Patch   | Balanced sampler         | $1.807 \pm 0.105$ | $0.302 \pm 0.009$ | $0.141 \pm 0.004$ |
| Patch   | ProGAN augmentation      | $2.337 \pm 0.118$ | $0.314 \pm 0.006$ | $0.151 \pm 0.003$ |
| Patch   | Focal                    | $1.156 \pm 0.043$ | $0.282 \pm 0.006$ | $0.118 \pm 0.002$ |
| WSI bag | RankMix                  | $0.407 \pm 0.041$ | $0.156 \pm 0.015$ | $0.022 \pm 0.003$ |
| WSI bag | MIL CE                   | $1.034 \pm 0.051$ | $0.195 \pm 0.009$ | $0.089 \pm 0.003$ |
| WSI bag | Balanced MIL             | $1.100 \pm 0.050$ | $0.201 \pm 0.009$ | $0.091 \pm 0.003$ |
| WSI bag | SC-MIL                   | $0.958 \pm 0.104$ | $0.205 \pm 0.013$ | $0.092 \pm 0.005$ |
| WSI bag | Weighted MIL             | $1.124 \pm 0.028$ | $0.210 \pm 0.011$ | $0.096 \pm 0.005$ |
| WSI bag | Focal MIL                | $0.763 \pm 0.064$ | $0.200 \pm 0.007$ | $0.078 \pm 0.006$ |

Table 9: Held-out test calibration metrics over three seeds. Lower is better for negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE). Values are not comparable across regimes because patch and WSI-bag models operate on different input units.

Table 9 applies the metrics in Section 4.1 and makes the discrimination–calibration tradeoff explicit. On patches, the method with the lowest ECE is not the method with the strongest macro F1, and the Scholz-style hybrid objectives improve class-balanced discrimination without improving likelihood-based calibration. ProGAN augmentation further weakens probability quality relative to the CE baseline. On WSI bags, RankMix is the only method that improves both class-balanced discrimination and calibration relative to plain MIL CE; the remaining MIL variants cluster near the baseline while trailing RankMix.

All WSI-bag calibration scores are computed from the same held-out test slides, hard slide labels, and raw softmax outputs of the final-epoch model. Every method uses the same evaluation routine: no temperature scaling, no validation-fit calibration, and no method-specific post-processing. An audit over stored test-result files confirms that all six WSI methods share identical test label sequences within each seed, that reported probabilities sum to one, and that stored NLL, Brier, and ECE values match recomputation from the saved probability matrix. The RankMix gap is therefore not an evaluation artifact: plain MIL CE is strongly overconfident (mean maximum predicted probability  $\approx 0.98$  versus  $\approx 0.91$  for RankMix), whereas RankMix’s soft mixing targets affect training only and not the held-out evaluation forward pass.

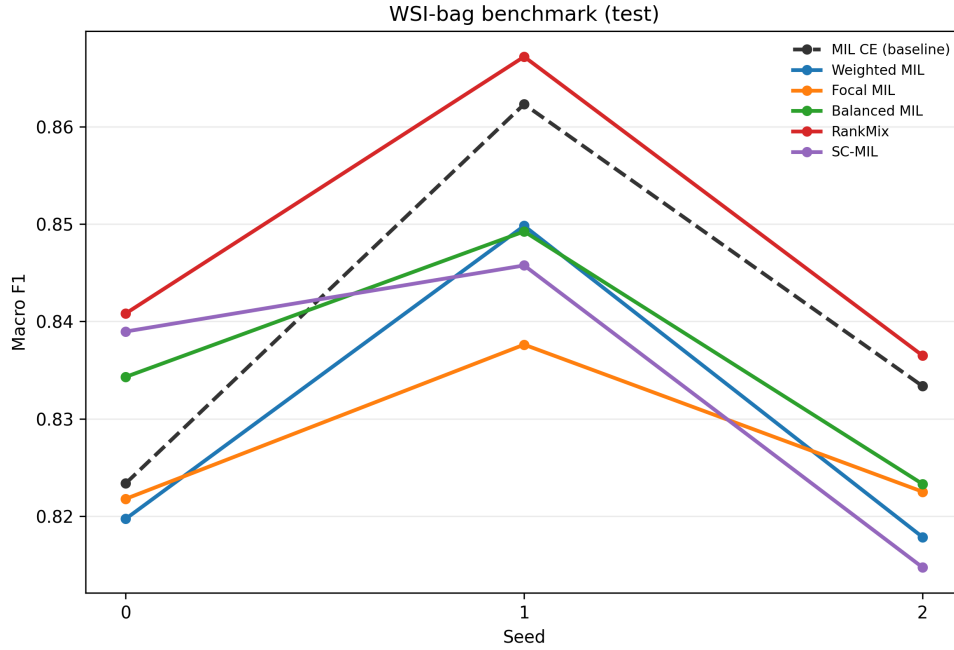


Figure 8: WSI-bag macro F1 on the held-out test split by random seed. The dashed line marks plain MIL CE; seed-wise spread shows why cross-method gaps should be read cautiously when standard deviations overlap (Table 6).

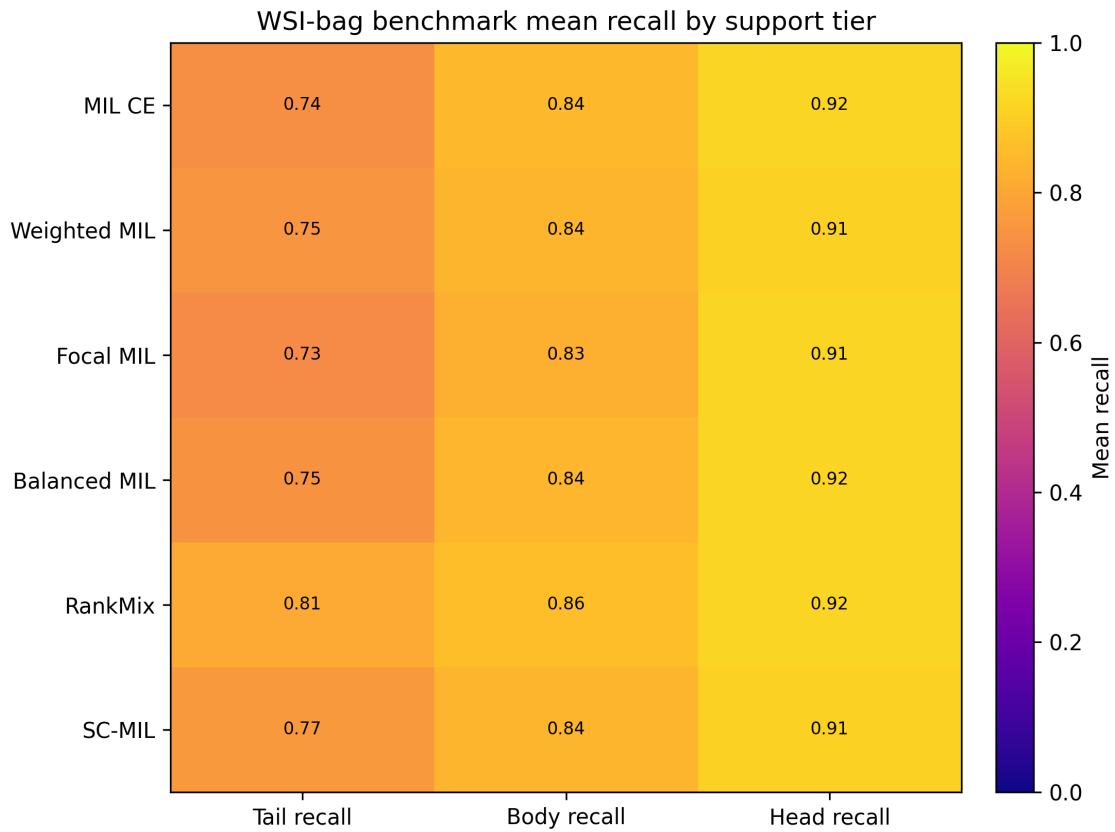


Figure 9: WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient for the long-tailed setting.

## 6 Discussion

The two benchmarks support different but compatible conclusions. In the patch benchmark, a strong frozen encoder compresses the performance spread: most methods cluster near macro F1 0.77–0.78, and the Scholz-style hybrid objectives provide small but consistent improvements over plain CE rather than large separations. ProGAN augmentation does not deliver the same advantage in this regime even though it adds large numbers of synthetic tail-class patches; the accompanying quality diagnostics suggest a combination of high Inception FID and substantial Virchow2 embedding-space separation from real training patches, rather than a simple failure to increase tail-class exposure. Focal loss is also unhelpful here, consistent with the interpretation that difficulty reweighting is redundant when the representation has already absorbed much of the class-separation problem. In the WSI-bag benchmark, RankMix shows the clearest benefit, especially in balanced accuracy, while macro-F1 gains remain modest; a WSI-native feature-mixing mechanism can still help over plain MIL CE even though the bags already use Virchow2 features.

These readings should be interpreted within the benchmark’s deliberate scope. The study is a controlled comparison rather than a full reproduction of every original paper: hyperparameters follow fixed configuration values and cited defaults rather than method-specific validation tuning, and both regimes use frozen Virchow2 features rather than end-to-end encoder adaptation. Rankings therefore describe how imbalance mechanisms behave when patch and slide classification are already supported by a strong pathology foundation model, not every published implementation under its preferred training recipe. The ProGAN adaptation likewise keeps the progressive-generator mechanism from Ruiz-Casado-style histology augmentation but applies it to multiclass TCGA-UT rather than reproducing the full GAN-family comparison from the source work.

The native long tail also means that class frequency is only one part of the difficulty signal. Some cancer types may be hard because few slides are available, while others may remain difficult because their frozen-feature distributions overlap with visually related classes. This distinction helps explain why pure exposure corrections have limited room to improve the patch benchmark: balanced sampling and class-weighted CE increase the influence of observed minority examples, but they do not add new within-class diversity or change the separability of the Virchow2 feature space. The same reasoning motivates reporting classwise recall and calibration alongside aggregate scores, because a method can improve average class exposure without resolving the hardest class-level decision boundaries.

Holding the encoder fixed within each benchmark therefore changes what an imbalance intervention can accomplish. On patches, the bottleneck shifts from learning stain and tissue morphology to calibrating decision boundaries under a long-tailed label distribution. Under that bottleneck, objectives that approximate macro F1 or MCC on balanced minibatches are better matched to the evaluation target than image-space synthesis or focal down-weighting of easy examples, but they can do so at the cost of worse likelihood scores without clear ECE improvements over CE. On slides, the bottleneck instead includes instance selection and bag aggregation, so RankMix can still add value by mixing informative subsequences even when the underlying features are strong, and in this benchmark it improves calibration as well as balanced accuracy.

This pattern reinforces the need to evaluate imbalance methods in their native input regimes and at the stage where the model is actually limited. A patch generator should not be judged by first converting synthetic images into one-instance MIL bags, and a MIL feature-mixing method should not be compared directly against a patch-level linear head on single embeddings. The shared split, encoder family, and 30-instance slide budget link the two benchmarks descriptively, but they do not license cross-regime inference: patch accuracy does not directly imply slide-level deployment quality, WSI-bag gains may reflect slide-native aggregation behavior absent

from the patch classifier, and neither absolute scores nor method orderings should be transferred across tables. The controlled design narrows the claims but makes them easier to defend: within each benchmark, the split, encoder family, head architecture, input budget, and evaluation code are shared. It also means that weak results should be read as weak results under the chosen fixed hyperparameters, not as proof that the mitigation families cannot work.

Several limitations remain. TCGA-UT patch labels come from pathologist-selected tumor regions; they are useful patch-level class labels but not dense annotations. The two benchmarks share slide-disjoint splits but differ in prediction unit and per-slide evidence budget, so their result tables remain non-transferable as discussed above. The experiments also use fixed, shared training budgets and method defaults rather than an exhaustive validation search over focal  $\gamma$ , class-weight scaling, sampler strength, hybrid-loss weighting, MIL hidden dimensions, RankMix mixing parameters, or ProGAN architecture and training schedule. Method-specific tuning could improve some metrics substantially; this study therefore supports a conservative claim about the tested fixed-protocol setting rather than a broad negative claim about imbalance mitigation.

## 7 Conclusion

TCGA-UT supports a cleaner imbalance study when patch-level and WSI-bag supervision are evaluated separately with regime-appropriate inputs. On controlled patch classification with frozen Virchow2 embeddings, Scholz-style CE plus soft F1 or soft MCC with balanced sampling provide small but consistent improvements over plain CE, whereas ProGAN augmentation and focal loss do not improve over that baseline. On WSI bags, RankMix shows the clearest benefit, especially in balanced accuracy, while macro-F1 gains remain modest. These two result sets are complementary rather than mutually confirming: patch-level metrics do not directly establish slide-level utility, WSI-bag metrics reflect slide-level aggregation under the same 30-instance slide budget, and methods should not be ranked across regimes. The practical implication is that imbalance mitigation should be matched to the supervision unit and to the learning stage that still limits performance; otherwise, an intervention can appear ineffective because the representation already absorbs the problem it was designed to solve, or because it was tested in the wrong regime.

## A Terminology

| Term                        | Meaning in this study                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------|
| <i>Slide / WSI</i>          | Diagnostic whole-slide image carrying one cancer-type label.                                        |
| <i>Patch</i>                | $256 \times 256$ H&E crop from a selected tumor region; patch labels inherit the slide cancer type. |
| <i>Patch embedding</i>      | Frozen Virchow2 vector for one controlled patch; input to the patch-level classifier head.          |
| <i>Region</i>               | Tumor area selected before patch extraction.                                                        |
| <i>Cancer type</i>          | The 32-way class label used for patch and slide-level classification.                               |
| <i>Feature chunk</i>        | Frozen Virchow2 feature representation used to assemble slide-level feature bags.                   |
| <i>WSI bag</i>              | Set of patch-feature instances for one slide, consumed by the MIL model.                            |
| <i>Head/body/tail class</i> | Fixed support tier from the full TCGA-UT slide distribution; see Section 3.8.                       |
| <i>Slide-disjoint split</i> | Split rule preventing any slide’s patches or features from crossing train, validation, and test.    |

Table 10: Terminology used for the two native input regimes (see Section 2.1).

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